

the start, but as long as you're learning, it's not cheating. Even professional software developers search the internet on a daily basis. In this section, you'll learn how to ask smart questions and search for answers on the internet.

When your program tries to carry out an invalid instruction, it displays an error message called a traceback. The traceback tells you what kind of error occurred and which line of code the error occurred on. Here's an example of a program that had an error while calculating how many slices of pizza each person should get:

```
Traceback (most recent call last):  
  File "pizza.py", line 5, in <module>  
    print('Each person gets', (slices / people), ' slices of pizza.')
```

ZeroDivisionError: division by zero

From this traceback, you might not realize that the problem is that the people variable is set to 0, and so the expression `slices / people` caused a zero-divide error. Error messages are often so short they're not even full sentences. Since programmers encounter them regularly, they're intended as reminders rather than full explanations. If you're encountering an error message for the first time, copying and pasting it into an internet search often returns a detailed explanation of what the error means and what its likely causes are.

If you're unable to find the solution to your problem by searching the internet, you can post your question to an online forum or email someone. To make this process as efficient as possible, ask specific, well-stated questions. This means providing full source code and error message details, explaining what you've already tried, and telling your helper what operating system and version of Python you're using. Not only will the posted answers solve your problem, but they can help future programmers who have your same question and find your post.

Typing Code

You don't have to be able to type fast to be a programmer, but it helps. While many people take a "hunt and peck" approach to typing, a faster typing speed can make programming less of a chore. As you work through the programs in this book, you'll want your eyes on the code and not on your keyboard.

There are free websites for learning how to type, such as <https://typingclub.com/> or <https://www.typing.com/>. A good typing program will display a keyboard and transparent hands on the screen so you can practice without the bad habit of looking down at the keyboard to find keys. Like every skill, typing is a matter of practice, and writing code can provide you with plenty of opportunities to type.

Keyboard shortcuts allow you to perform actions in a fraction of the time it takes to move the mouse to a menu and perform the action. A shortcut is often written like "CTRL-C," which means pressing down one of the